Reciprocity and return
Completed experiments

FINAL EXPERIMENT REPORT | 11 OCTOBER 2026 | EMM PROJECT

This report consolidates the completed analyses of reciprocity and next-month return in Reddit reply networks. The main search and robustness comparisons are complete for the implemented activity-based specification. The results describe associations and are not estimates of causal effects.

Analysis	Status	Completed scope
Supporting analyses	Complete	Temporal and reciprocity EDA; pooled model; 2,046 community models
M: reference discovery	Complete	Activity selectors; 29 valid groups; top 10 reported
A1: measure	Complete	Binary versus proportional; overlap, signs and reference differences
A1: weighted variant	Not applicable	Full raw-data audit found no repeated-reply intensity
A2: dependence	Complete	Ordinary versus user-clustered inference; reference plus fixed top 10
A3: effect scale	Complete	Binary AMEs; reference plus fixed top 10; ranking and direction checks

What the experiments establish

1. Adjusted reciprocity-return associations are negative in the reference and all 29 activity subgroups evaluated in M.

2. Nine of the top ten contexts survive the change from binary to proportional reciprocity. Some reference-relative conclusions remain measure-sensitive.

3. User clustering modestly widens uncertainty. Probability-scale analysis changes the ordering of the strongest subgroup differences.

Scope: months 1-10; activity selectors only; no lagged controls or community-context selector search. The weighted variant is unavailable in the supplied data. See pages 8-9 for limits and reproduction.

1. Data, model and supporting analyses

The analysis contains 36,031,393 user-subreddit-month observations from 3,354,982 users across 2,046 subreddits. The next-month return rate is 36.39%. Eleven monthly networks provide ten observed transitions; month 11 supplies the last return label.

Variables and fitted model

Return means appearing as a reply author in the same subreddit in the following month. Binary reciprocity is 1 when reciprocal_partners > 0. Proportional reciprocity is reciprocal_partners / outgoing_partners; this definition was checked on every analysis row.

The logistic model includes reciprocity, ln(1 + replies_sent), ln(1 + replies_received), and month and subreddit fixed effects. Subgroup models use the same specification within each subgroup. The global reference uses the full analysis population, including the subgroup.

Sent and Received throughout this report refer to the existing replies_sent and replies_received fields. The raw adjacency lists contain no repeated targets, so these supplied network counts do not retain repeated-message intensity.

Completed foundation

Temporal EDA found predominantly short-lived or intermittent participation. Across subreddits, extending the return window from one to three months increased return rates by an unweighted mean of 14.34 percentage points. This EDA does not change the one-month outcome used in M and A1-A3.

Unadjusted return rates were 44.37% with binary reciprocity and 27.66% without it, a 16.71-percentage-point difference. After adjustment, the pooled coefficient was -0.2413, with clustered 95% CI [-0.2435, -0.2390]; the odds ratio was 0.786 and the binary AME was -4.44 percentage points. Raw group differences and adjusted model estimates represent different comparisons.

Separate community models converged for all 2,046 subreddits. Coefficients were negative in 2,021 and positive in 25; nominal 95% intervals lay below zero in 1,782 and above zero in three. The remaining 261 included zero. The median AME was -6.33 percentage points (IQR -8.25 to -4.55). These comparisons were exploratory and unadjusted for multiple testing.

Sources: ../network_eda/; ../eda_outputs/figures/; ../logistic_resgression/pooled_logistic.ipynb; ../logistic_resgression/results/subreddit_logistic_results.csv; A2 and A3 reference checkpoints.

2. M: reference subgroup discovery

The search uses Sent thresholds 1 and 3 and Received thresholds 0, 1 and 3, with <= and > conditions. Beam width is 10, maximum depth is 2, and minimum support is 5,000. Reporting k = 10 is distinct from beam width. The quality score is sqrt(n) x |beta_subgroup - beta_reference|.

The binary search returned 29 valid subgroups: nine with one condition and twenty with two. All 29 reciprocity coefficients were negative, ranging from -0.522 to -0.177, against a global reference of -0.2413. The highest-quality subgroup was Sent > 3 and Received > 3 (n = 5,056,348; return rate 75.78%; beta = -0.522; 95% CI [-0.537, -0.507]). Its AME was -7.55 percentage points.

M rank	Subgroup	n	Return %	Beta [95% CI]	Signed delta	Quality
1	Sent>3 and Received>3	5,056,348	75.78	-0.522 [-0.537, -0.507]	-0.281	632.0
2	Sent>1 and Received>1	11,289,290	59.58	-0.429 [-0.433, -0.424]	-0.187	629.1
3	Sent>3 and Received>1	6,448,956	72.37	-0.487 [-0.497, -0.477]	-0.246	624.7
4	Sent>1 and Received>0	14,380,085	55.42	-0.400 [-0.404, -0.396]	-0.159	601.7
5	Sent>3 and Received>0	6,871,536	71.45	-0.442 [-0.450, -0.434]	-0.201	525.6
6	Sent>1 and Received<=3	9,638,277	42.97	-0.398 [-0.402, -0.394]	-0.157	486.2
7	Sent>1	16,170,924	53.71	-0.362 [-0.365, -0.358]	-0.120	484.1
8	Sent>1 and Received>3	6,532,647	69.57	-0.406 [-0.414, -0.398]	-0.164	420.3
9	Sent>3	7,031,072	71.09	-0.379 [-0.386, -0.371]	-0.137	364.4
10	Received<=3 and Sent<=3	26,881,694	26.73	-0.309 [-0.312, -0.307]	-0.068	353.5

Subgroups overlap; their sizes must not be added. CIs use user-clustered standard errors. Signed delta = subgroup beta minus reference beta. The original CSV field delta is the absolute difference used in quality. More negative signed delta means a stronger negative conditional association.

The weakest negative coefficient among all 29 valid groups was -0.177 for Received>3 and Sent<=1 (n = 642,328). 28/29 intervals exclude the reference point. Magnitudes and probability-scale effects are more informative than this diagnostic alone.

Sources: ../beam_search/emm_beam_search_results.csv and M_reference_discovery/tables/top_10_subgroups.csv. AME values are supplied by A3. The subgroup definitions use the same activity variables included as regression controls.

3. A1: reciprocity measure

Binary and proportional reciprocity retained nine of the ten highest-quality subgroup descriptions (Jaccard overlap 9/11 = 0.818). All nine shared top-ten subgroups had negative coefficients and more negative coefficients than their own measure-specific references. Across all 29 shared groups, coefficient signs agreed in 29/29 cases, while the direction of the subgroup-reference difference agreed in 21/29. Thus the main selected contexts and association signs were broadly stable, but ranking and some reference-relative conclusions depended on the definition of reciprocity.

Proportional reference: beta -0.3728, clustered SE 0.001347, 95% CI [-0.3755, -0.3702]. The same search returned 29 valid groups from 34 attempted descriptions; five lacked reciprocity variation.

Context	Binary rank	Prop. rank	Binary beta	Prop. beta
Sent>3 and Received>3	1	4	-0.522	-1.440
Sent>1 and Received>1	2	7	-0.429	-0.900
Sent>3 and Received>1	3	3	-0.487	-1.367
Sent>1 and Received>0	4	6	-0.400	-0.859
Sent>3 and Received>0	5	1	-0.442	-1.339
Sent>1 and Received<=3	6	10	-0.398	-0.786
Sent>1	7	5	-0.362	-0.838
Sent>1 and Received>3	8	8	-0.406	-1.026
Sent>3	9	2	-0.379	-1.327
Received<=3 and Sent<=3	10	23	-0.309	-0.331
Sent>3 and Received<=3	20	9	-0.371	-1.344

The table is the union of both top-ten lists; ranks above 10 identify membership changes. Sent > 3 and Received <= 3 enters (20 to 9); Sent <= 3 and Received <= 3 leaves (10 to 23). Betas represent different reciprocity contrasts and are not directly comparable probability changes.

Weighted variant: not applicable to the supplied data

The audit examined all 2,046 JSON files, 22,506 monthly networks and 136,829,959 target entries across 40,050,308 outgoing user-month lists. No list contained repeated targets. A distinct repeated-reply-weighted experiment cannot be recovered from these inputs. This does not imply that users never replied repeatedly in the underlying Reddit activity.

Six proportional subgroups had separation exclusions; binary refits verified the same excluded rows and reproduced saved coefficients and SEs. One of these groups enters the proportional top ten. Input n remains the quality-function support; fitted_rows records actual estimation size. Sources: A1_reciprocity_measure/tables/ and audits/.

4. A2: dependence between observations

The global reference and fixed binary top ten were refitted once each using the original likelihood and controls. User-clustered CRV1 uncertainty was reproduced, then ordinary model-based (iid) uncertainty was calculated on the same fitted model. All eleven models converged and retained their complete input samples.

User clustering increased the reference standard error by 13.1% and top-ten subgroup standard errors by 2.1%-10.8% (median 5.9%). All ten subgroup confidence intervals remained below zero under both approaches. Coefficients and the quality ranking were unchanged. M already used user-clustered inference; A2 supplies its explicit comparison with ordinary model-based uncertainty.

M rank	Ordinary SE	Clustered SE	Change %	Ordinary 95% CI	Clustered 95% CI
Ref	0.001014	0.001146	13.1	[-0.2433, -0.2393]	[-0.2435, -0.2390]
1	0.007484	0.007645	2.1	[-0.5370, -0.5077]	[-0.5373, -0.5074]
2	0.002310	0.002435	5.4	[-0.4330, -0.4240]	[-0.4333, -0.4237]
3	0.004860	0.005036	3.6	[-0.4968, -0.4777]	[-0.4971, -0.4774]
4	0.001717	0.001847	7.6	[-0.4033, -0.3966]	[-0.4035, -0.3963]
5	0.003967	0.004166	5.0	[-0.4495, -0.4340]	[-0.4499, -0.4336]
6	0.001842	0.001961	6.5	[-0.4015, -0.3943]	[-0.4017, -0.3940]
7	0.001520	0.001677	10.3	[-0.3646, -0.3587]	[-0.3649, -0.3584]
8	0.004055	0.004181	3.1	[-0.4137, -0.3978]	[-0.4139, -0.3975]
9	0.003505	0.003883	10.8	[-0.3856, -0.3718]	[-0.3863, -0.3711]
10	0.001223	0.001312	7.3	[-0.3119, -0.3071]	[-0.3120, -0.3069]

M ranks refer to the subgroup descriptions on page 3. SE change = 100 x (clustered SE / ordinary SE - 1). Both sets of intervals are pointwise 95% intervals. All subgroup intervals remain below the reference coefficient under both methods.

Interpretation

Allowing dependence within users modestly widens uncertainty without overturning the negative-association evidence in these selected groups. It is expected that coefficient signs, AME point estimates and quality rankings do not change when only the covariance calculation changes. This comparison does not isolate dependence from every possible source of model misspecification.

A subgroup confidence interval excluding the global reference coefficient is a descriptive diagnostic, not a formal test of the difference between two estimated coefficients. The reference includes the subgroup, so a formal test must account for their joint uncertainty. Intervals are pointwise and have no multiple-testing or post-selection adjustment.

Sources: A2_user_dependence/tables/a2_comparison.csv; reports/summary.json; audits/validation.json. All ten clustered subgroup fits reproduce M; the reference agrees with the saved pooled notebook. Clustered results remain the basis for interpretation.

5. A3: effects on the probability scale

For binary reciprocity, the AME is the average discrete change in predicted return probability when reciprocity changes from 0 to 1, holding the other model inputs fixed for each observation. An AME of -7.55 percentage points means an average model-predicted reduction of 7.55 points for that contrast; it is not a 7.55% relative reduction and is not a causal estimate.

The reference AME was -4.44 percentage points. AMEs for the fixed top ten ranged from -9.01 to -5.59 percentage points. All ten subgroup-reference differences had the same direction on the coefficient and probability scales. However, the Spearman correlation between rankings by absolute coefficient difference and absolute AME difference was only 0.370. Directional conclusions were consistent, while the ranking of substantive effect magnitudes depended on the scale.

M rank	Signed delta beta	AME (pp)	Delta AME (pp)	|Delta beta| rank	|Delta AME| rank
1	-0.281	-7.55	-3.12	1	5
2	-0.187	-8.36	-3.92	4	2
3	-0.246	-7.83	-3.40	2	4
4	-0.159	-8.11	-3.67	6	3
5	-0.201	-7.33	-2.90	3	7
6	-0.157	-9.01	-4.57	7	1
7	-0.120	-7.41	-2.98	9	6
8	-0.164	-6.87	-2.43	5	8
9	-0.137	-6.42	-1.98	8	9
10	-0.068	-5.59	-1.15	10	10

M rank is the original size-weighted quality rank. The last two columns rank absolute subgroup-reference differences without size weighting. Reference AME: -4.44 percentage points. Subgroups are identified on page 3.

Which context has the largest probability difference?

The most negative AME is -9.01 percentage points for Sent>1 and Received<=3 (M rank 6), a difference of -4.57 points from the reference. The strongest coefficient context, Sent > 3 and Received > 3, has AME -7.55 points and ranks fifth by absolute AME difference.

Each AME is averaged over its own fitted population. Differences therefore reflect both model estimates and covariate composition, rather than a common standardized population. A3 provides point estimates only: no AME confidence intervals or post-selection uncertainty were estimated. Coefficient/AME sign agreement is expected in this binary main-effects specification and is not an independent robustness test.

Sources: A3_probability_scale/tables/top_10_ame_comparison.csv, reports/summary.json and model_checkpoints/. Eleven refits reproduced M's coefficients to numerical precision. Ordinary covariance was used for those point-estimate refits; original clustered coefficient CIs were retained.

6. Visual comparison of the results

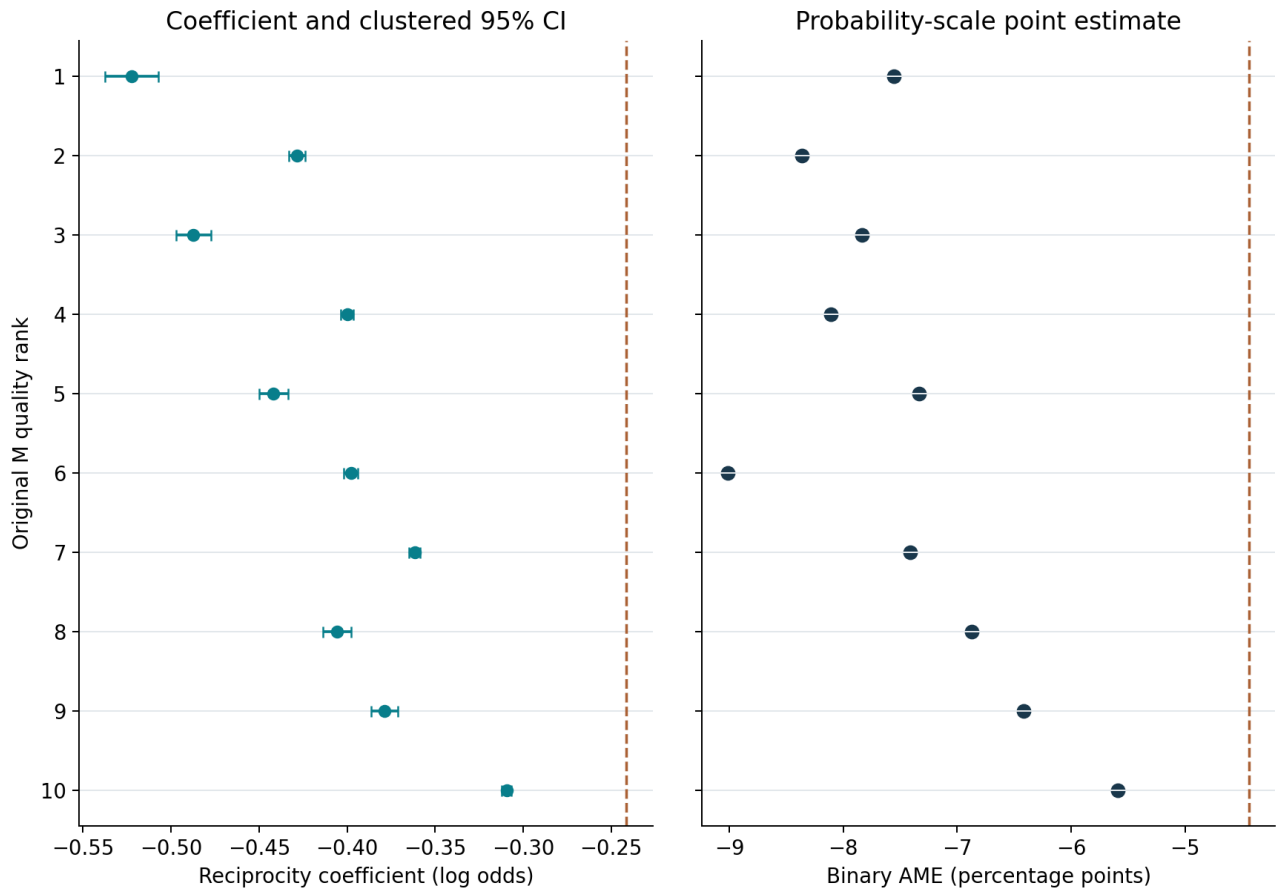


Figure 1. The same ten subgroups, indexed by M rank. Left: coefficients and user-clustered 95% intervals; the dashed line is the global coefficient. Right: binary AME point estimates; the dashed line is the global AME. AME uncertainty was not estimated. Separate horizontal scales are intentional.

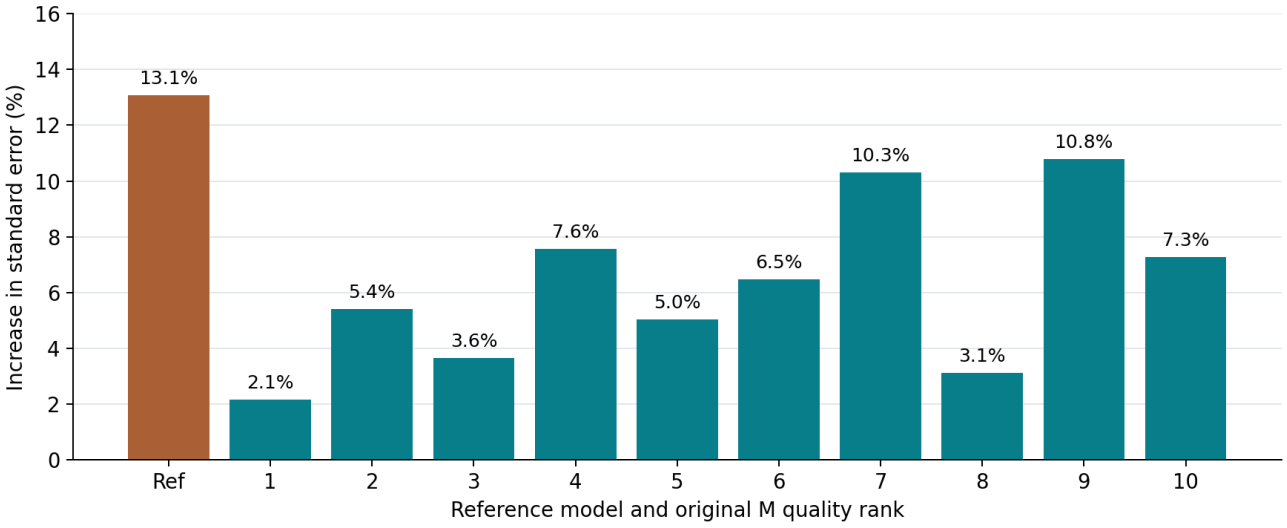


Figure 2. Percentage increase in SE after user clustering. Ref denotes the global model; 1-10 are M ranks. Both figures use the canonical results tables and can be regenerated without model refits.

7. Interpretation and remaining scope

Overall interpretation

The positive raw reciprocity-return difference does not persist after adjustment for activity and fixed effects. Negative adjusted associations appear across the activity contexts evaluated. The main selected contexts are broadly robust to binary versus proportional measurement and to clustered uncertainty, while the order of effect magnitudes is sensitive to whether coefficients or probability differences are used.

Substantive interpretation should report both the coefficient contrast and the AME. A more negative logistic coefficient does not necessarily imply the largest average probability difference. The AME analysis supports directional heterogeneity across the selected activity contexts, but the modest rank correlation argues against treating the coefficient-based ordering as the sole substantive result.

Limits of the completed experiments

Activity context: subgroup selectors are Sent and Received, which are also regression controls. These results describe variation across activity levels; the search did not include separate community-context variables. Community-specific regressions are supporting analyses, not a community-selector beam search.

Model scope: fitted transitions are 1-10, with contemporaneous activity controls and month/subreddit fixed effects. Lagged controls and the corresponding transitions 2-10 specification have not been run.

A subgroup confidence interval excluding the global reference coefficient is a descriptive diagnostic, not a formal test of the difference between two estimated coefficients. The reference includes the subgroup, so a formal test must account for their joint uncertainty. Intervals are pointwise and have no multiple-testing or post-selection adjustment.

Effect scale and data: AMEs use each group's own covariate distribution and have no uncertainty intervals. Weighted repeated-reply intensity is unavailable. All results remain descriptive; none establishes that reciprocity causes lower return.

What remains for the paper

The experiment report, tables and paste-ready results text are complete. Align the methods with the implemented activity-only, non-lagged specification; if broader selectors or lagged controls remain a requirement, those are additional experiments. Describe weighted A1 as not applicable, retain clustered CIs, explain the two effect rankings, and distinguish interval exclusion of the reference point from a formal difference test.

manuscript_results.txt and manuscript_results.tex contain results prose for M and A1-A3. No manuscript source was edited, and no new experiments were run during packaging. The existing LaTeX tables are included as exported; their layout has not been compiled in the target manuscript.

8. Files and reproduction

Each code or result file has one canonical location. Added experiment code and outputs live under experiments/. Original search, EDA and regression code remains in its existing project folders. Duplicate copies have been removed; reproducibility/manifest.json records the current paths and SHA-256 hashes.

Folder / file	Purpose
README.txt	Start here: folder map, commands and dependencies
M_reference_discovery/	Added reporter and top-ten tables; original search stays in ../beam_search/
A1_reciprocity_measure/	Measure comparison, weighted audit, results, checkpoints and single data cache
A2_user_dependence/	Dependence runner, SE/CI results and checkpoints
A3_probability_scale/	AME runner, effect/ranking results and checkpoints
supporting_analyses/	Guide to existing EDA and model files; no duplicate source copies
figures/	Report figures in PNG and vector SVG
code/	Report builders, shared path routing and experiment launcher
reproducibility/	Current file manifest, environment, source checks and relocation audit

Rerunning

From the project root, python experiments/code/run_experiment.py A2 prints the command. Add --execute to run it. Choices: M, M-report, A1, A1-weighted-audit, A2 and A3. Added scripts can also run directly at their documented paths. The launcher preserves the original M entry point. Use the recorded analysis environment, including pyfixest 0.60.0.

Generated outputs go directly to tables/, reports/, latex/, audits/ and model_checkpoints/ inside each experiment. A1/A2 share the single parquet cache under A1_reciprocity_measure/data_cache/. The analysis CSV and raw networks remain in their original locations. Completed checkpoint signatures were migrated after verifying that only paths and output routing changed; numerical estimates were preserved.

To rebuild the report without refitting, run code/build_bundle.py using the analysis Python, then code/render_report.py using a Python with reportlab. The builder reads canonical files and does not copy them. Reruns of experiments may update their generated results; they do not edit original source code.

Original evidence: ../beam_search/emm_beam_search_results.csv, ../network_eda/, ../eda_outputs/figures/ and ../logistic_resgression/. Source references in this report are relative to experiments/. Historical audits retain paths as they existed at the time; consolidation_plan.json maps those paths to current locations.